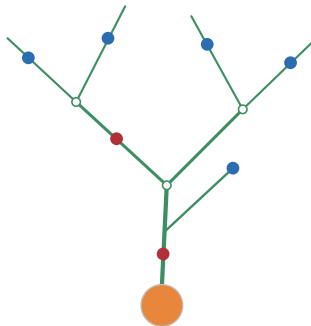

What dendritic structure adds to learning

Credit assignment from neurons to branches: coordinates, subtree addresses and route gain

Houman Safaai · Maceo Richards · Bernardo L. Sabatini
Kempner Institute, Harvard University · Harvard Medical School

Kempner Learning Dynamics Workshop · 2026



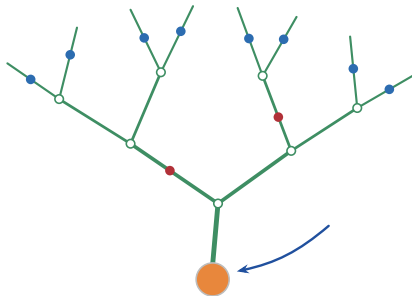
One error arrives. Ten thousand synapses must decide.

$\sim 10^4$

synapses arrayed on one branching arbor

Which of them should change, in which direction, by how much?

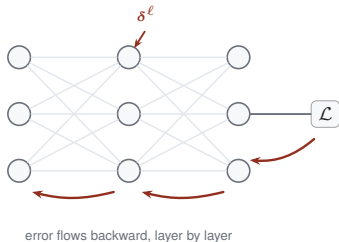
The task delivers one output error — at the soma.



one output error arrives at the soma

The tree between soma and synapse is either in the way — or part of the answer.

Learning dynamics reduce to who gets credit



GRADIENT FLOW

$$\frac{d\theta}{dt} = -\eta \nabla_{\theta} \mathcal{L}$$

$$\frac{\partial \mathcal{L}}{\partial W_{ji}^{\ell}} = \underbrace{h_i^{\ell-1}}_{\text{presynaptic input}} \underbrace{\delta_j^{\ell}}_{\text{destination error}}$$

- **SIGN** potentiate or depress
- **MAGNITUDE** how strongly this parameter moves the loss
- **DESTINATION** every parameter indexed exactly

A theory of biological learning must say what stands in for δ at each synapse.

Backprop is the reference, not the mechanism

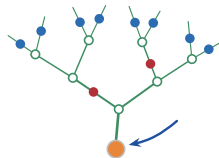
BACKPROP UPDATE

$$\Delta W_{ji}^{\ell} = -\eta h_i^{\ell-1} \delta_j^{\ell}, \quad \delta^{\ell} = \underbrace{(W^{\ell+1})^T \delta^{\ell+1}}_{\text{weight transport}} \odot f'(a^{\ell})$$

what backprop demands

- TRANSPORT
- COORDINATION
- NONLOCAL STATE

a synapse sees: input, compartment state, one teaching signal.

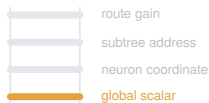


Judge every biological rule by the information it puts in place of δ .

ACT II

Which neuron should change?

The state of the art, read analytically



Four families of answers

■ SCALAR MODULATOR

$\Delta w_i \propto e_i m_u$
Frémaux & Gerstner 2016

■ STATE INFERENCE

teaching signal = nudged-state difference
Scellier & Bengio 2017; Whittington & Bogacz 2017

■ RANDOM FEEDBACK COORDINATE

$\hat{\delta}^\ell = B^\ell \delta^{\text{out}}$
Lillicrap et al. 2016; Nøkland 2016

■ DENDRITIC ERROR COMPARTMENTS

one error per neuron, held beside its activity
Guerguiev et al. 2017; Sacramento et al. 2018; Payeur et al. 2021; Song et al. 2024

Each family delivers a different error to the neuron; none indexes a branch.

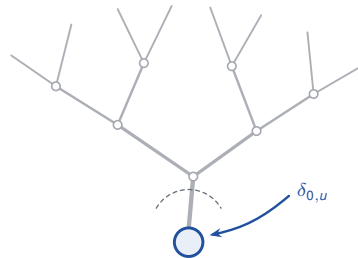
They converge on one per-neuron coordinate δ_u

point neuron: $s_u = \sum_i w_i x_i$, $V_u = f(s_u)$

POINT NEURON

$$\frac{\partial \mathcal{L}}{\partial w_i} = \underbrace{x_i f'(s_u)}_{\text{synapse-local eligibility}} \underbrace{\delta_u}_{\text{one shared coordinate}}$$

No variable in this equation can address one subtree differently from another.

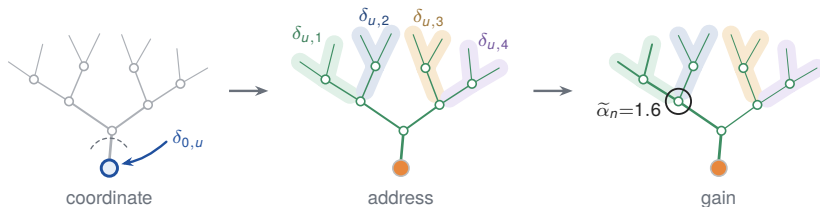


Point-based theories have no addressable variable inside the arbor.

This talk: what does the tree add after $\delta_{0,u}$ arrives?

PREVIEW

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{e_i}_{\text{local state}} \underbrace{\delta_{0,u}}_{\text{neuron coordinate}} \underbrace{\tilde{\alpha}_{n(i)}}_{\text{address and route gain}}$$

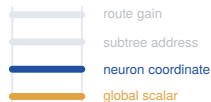


Three candidate roles for the tree: local state, subtree addresses, route gains.

ACT III

What exactly does a tree add to a teaching signal?

Theory: eligibility, transport, capture, bandwidth and gain



The forward model is a conductance quotient

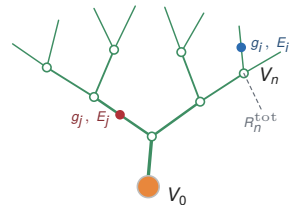
MODEL

$$V_n = \underbrace{R_n^{\text{tot}}}_{\text{conductance quotient}} \left(\sum_{i \in \mathcal{S}_n} g_i x_i E_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c + g_n^L E_L \right)$$

$$R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}, \quad g_n^{\text{tot}} = g_n^L + \sum_{i \in \mathcal{S}_n} g_i x_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}$$

Every input raises both the weighted drive and the total conductance.

Journal Eq. 5 (steady-state compartment equation); $a_c = f_c(V_c)$ propagates rootward.



state carried at every compartment

The quotient is the mechanism: normalization lives in the denominator.

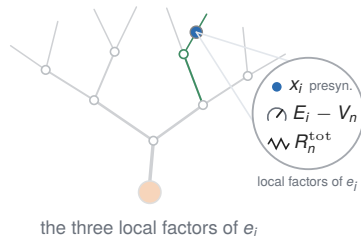
The quotient rule yields an exact local eligibility

EXACT LOCAL ELIGIBILITY

$$\frac{\partial V_n}{\partial g_i} = \underbrace{x_i}_{\text{presynaptic activity}} \underbrace{R_n^{\text{tot}}}_{\text{input resistance}} \underbrace{(E_i - V_n)}_{\text{driving force}} \equiv e_i$$

The quotient rule creates the driving-force factor; additive-current models lose it.

Journal Eq. 6; full quotient-rule steps in backup.



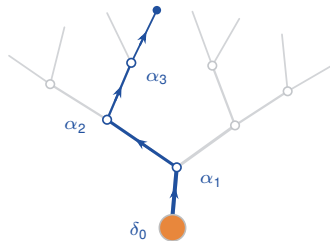
Conductance normalization does not destroy locality; it creates the exact factor $E_i - V_n$.

Every exact gradient factors: eligibility $\times$ transported error

THEOREM

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{e_i}_{\text{local eligibility}} \underbrace{\delta_n}_{\text{transported error}}, \quad \delta_n = \delta_{0,u} \tilde{\alpha}_n$$

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}$$

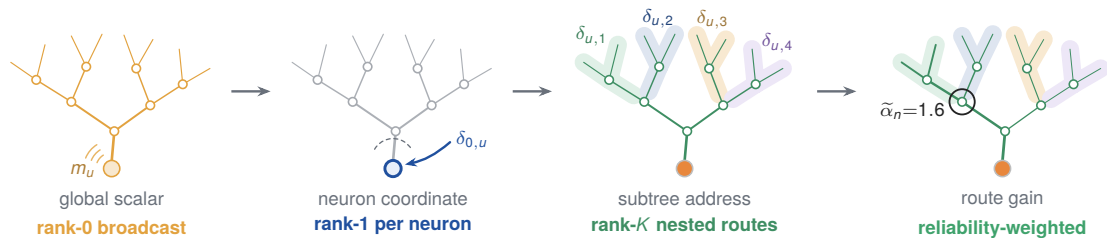


one route from soma to every site

Journal Eqs. 2–4 and 6–7; exact reconstruction matched autodiff on every diagnostic run.

Every biological proposal is an explicit approximation to the transported factor δ_n .

Dendritic credit is a four-level hierarchy of error fields



This ladder is the map of the talk: each experiment tests one rung against the next.

A stochastic credit operator predicts when routes help

update $\mathbf{w}^+ = \mathbf{w} - \eta M \hat{\mathbf{g}}$, teaching field $\hat{\mathbf{g}} = \mathbf{g} + \boldsymbol{\xi}$, noise covariance Σ

DESCENT BOUND AND PHASE UTILITY

$$\mathbb{E} \mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \underbrace{\mathbf{g}^\top M \mathbf{g}}_{\text{retained signal}} + \frac{L\eta^2}{2} \left(\underbrace{\|M\mathbf{g}\|_2^2}_{\text{curvature bias}} + \underbrace{\text{tr}(M\Sigma M^\top)}_{\text{admitted noise}} \right)$$

$$U(M) = \frac{[\mathbf{g}^\top M \mathbf{g}]^2}{2L \{ \|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top) \}}, \quad \mathbf{g}^\top M \mathbf{g} > 0$$

Positivity is the feedback-alignment condition; the bound is exact for the matched quadratic. Journal Eqs. 8–9.

A restricted route helps exactly when retained signal beats bias plus admitted noise.

Capture and the tree spectrum pick the best dictionary

CAPTURE

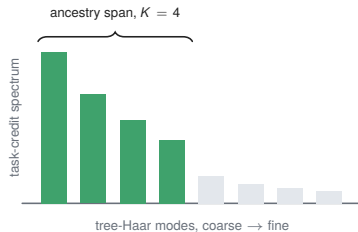
$$C_K(\mathbf{g}; \Phi, W) = \frac{\|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}$$

task-gradient energy inside the anatomical span

spectral form: $C_K(\bar{C}_g) = \text{tr}(\bar{P}_\Phi \bar{C}_g) / \text{tr}(\bar{C}_g)$; ancestry spans the K coarsest tree-Haar modes

+0.55

more captured task energy than a random rank-matched route —
ancestry at $K=4$ under full task-tree alignment, 50/50 seeds



coarse-dominant spectrum $\Rightarrow$ the anatomical span is
Ky-Fan optimal

Anatomy is optimal when task covariance is coarse-dominant in the tree's own basis.

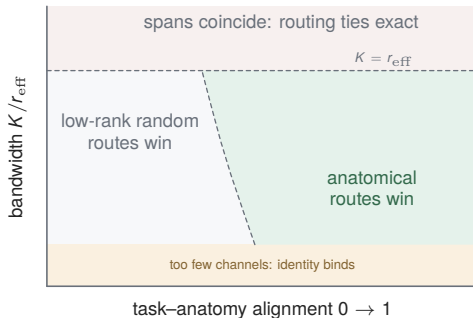
Bandwidth has an interior optimum

PROJECTION BEATS FULL BANDWIDTH IFF

$$\underbrace{\text{tr}[(I - P)\Sigma]}_{\text{rejected noise}} > \underbrace{\|(I - P)\mathbf{g}\|_2^2}_{\text{discarded signal}}$$

identity-curvature quadratic at $\eta = 1/L$

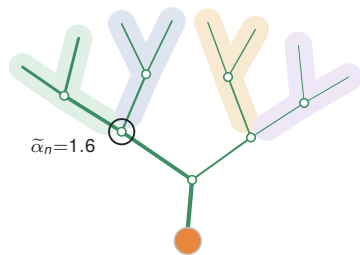
Bandwidth pays only while each added route scale brings more signal than admitted noise.



conceptual regime bands, not fitted — the measured plane closes the talk

This phase plane sets up every experiment in the next act.

Conductance implements reliability-weighted route gain



stroke width = route gain

OPTIMAL BRANCH ATTENUATION

$$a_b^* = \min \left\{ 1, \underbrace{S_b}_{\text{branch signal}} / \underbrace{c(S_b + N_b)}_{\text{dose} \times (\text{signal} + \text{noise})} \right\}$$

A shunt is a rank-one conductance edit of the adjoint (Sherman–Morrison); dose raises its amplitude monotonically, then saturates.

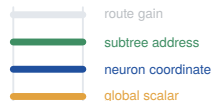
Journal, credit-operator section; rank-one identity and selectivity criterion in backup.

Branch-local conductance can realize the optimal attenuation a_b^* — physically and locally.

ACT IV

Do subtree addresses and route gains actually pay?

A battery of matched experiments, one rung at a time



Every comparison is matched-resource and paired by seed

- **MATCHED BUDGETS**

only credit information changes

- **PRESPECIFIED DESIGNS**

outcome-independent validity rules

- **PAIRED SEEDS**

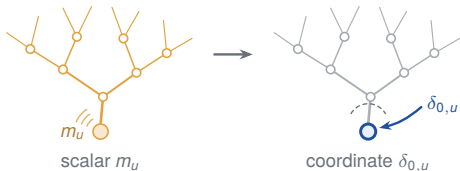
effects are within-seed differences

- **POINT EMULATION**

same routed fields $\Rightarrow$ same result, by construction

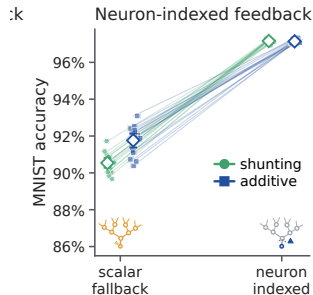
What follows are resource claims about structured feedback, under matched forward models.

Per-neuron bandwidth dominates



+4.9 to +14 pp

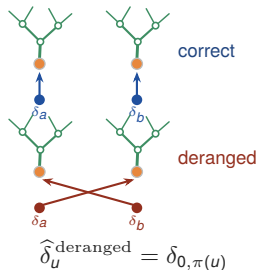
one signed coordinate per neuron replaces one broadcast scalar; 120/120 valid pairs



15-seed regular-tree replication shown; the range and 120/120 pairs are the prospective 640-run cohort

Per-neuron bandwidth, not routing — the classic bottleneck (Werfel–Xie–Seung 2005).

Correct ownership adds a small, reproducible margin

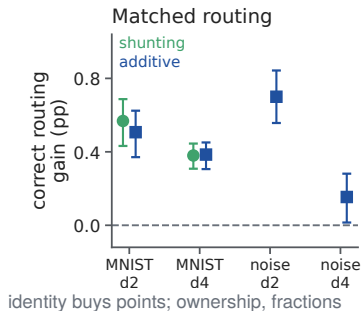


+0.15 to +0.70 pp

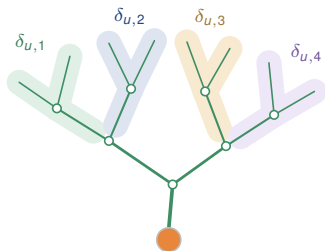
correct neuron-to-tree map, favored in 58/60 valid pairs

— same values, same bandwidth

Identity is mostly bandwidth; owning the right tree adds a small, real margin.



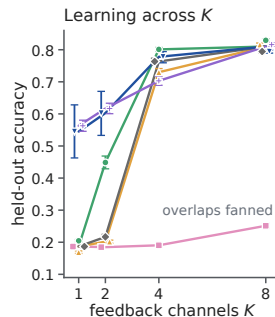
Subtree addresses win only at the predicted budget



K nested route fields

+1.3 pp at $K=4$

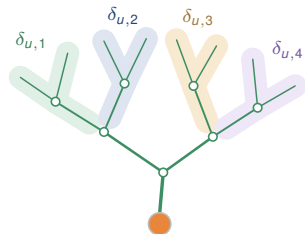
only at the predicted budget (15/20 seeds,
 $P=0.0048$); 2,700 matched fits



$K=1, 2$ lose, $K=8$ ties — the crossover lands where the theory put it

An interior optimum in feedback bandwidth — observed where predicted.

Depth pays when route scales match the task hierarchy

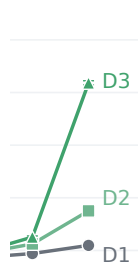


signal at levels $\leq H$, noise below

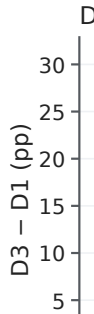
+27.7 pp

final step of the alignment dose (D3–D1); optimal routed depth = H for every H

se



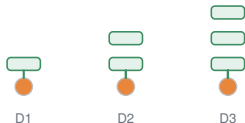
N



controlled validation under an imposed hierarchy — not a discovery about arbitrary tasks

Alignment is a steep boundary the data cross — not a linear dose.

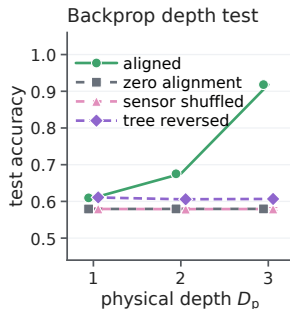
Aligned divisive depth helps; depth per se costs



$$V_n = \frac{E_n + C_n}{1 + E_n + I_n + G_n + \epsilon}$$

+31 pp

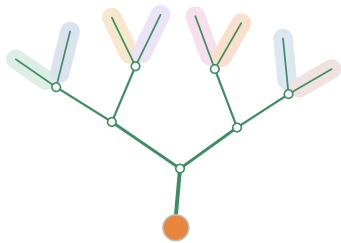
aligned D3–D1, constructed; controls ≈ 0 ; depth alone costs (40/40)



same branch units and parameters at every depth; aligned composition rises, every control stays flat

A constructed existence proof — conditional on alignment, never free.

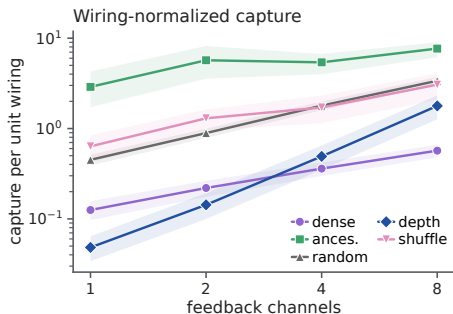
Real arbors are sparse, coarse route dictionaries



ancestry dictionary, 8 channels

2.7×

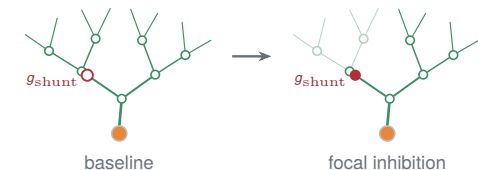
anatomy-specific capture per wiring, beyond a density-matched shuffle — 8 MICrONS cells



eight sparse ancestry channels reach most of dense capture at 7% of its wiring

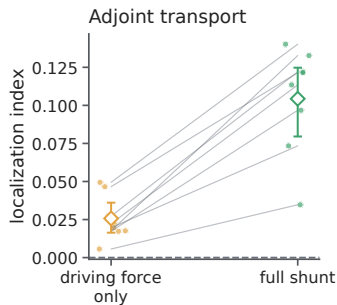
Sparsity does most of the work; anatomy adds a bounded margin on top.

A focal shunt rewrites transport below the branch



0.069

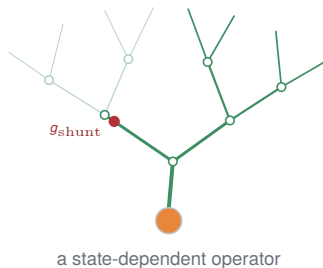
shunt – additive localization contrast; 8/8 cells,
dose-monotone — soma and output error matched



freezing the adjoint kills the effect; restoring it recovers it

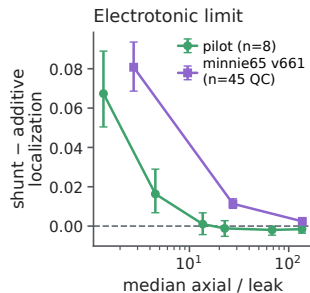
Shunting edits the transported error, not the local driving force.

The mechanism needs high-conductance states



0.067

contrast at in-vivo-like R_m (8/8 cells); null at textbook resting calibration



active Na/K/Ca/HCN/NMDA ensembles show the same contrast — attenuation only, no sign flips

State-dependence is the mechanism claim: a testable boundary, not a failure.

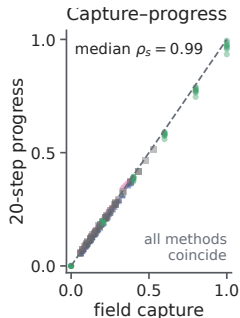
Anatomy alone is not enough; alignment rescues it

$$\mathbf{g}(a) = \sqrt{a} \mathbf{u}_{\parallel} + \sqrt{1-a} \mathbf{u}_{\perp}, \quad \|Q Q^T \mathbf{g}(a)\|^2 = a$$

Measured responses give a null — a boundary, not a failure. Rotating alignment rescues the same dictionary.

$$\rho_s = 0.99$$

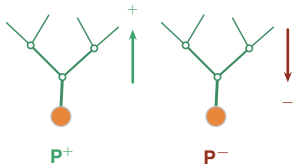
capture predicts 20-step learning; $a=0$ loses 8/8, $a=1$ wins 8/8



same cells, same channels, fixed gradient energy and curvature; only alignment rotates

Alignment, not anatomy alone, determines utility.

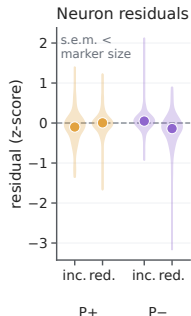
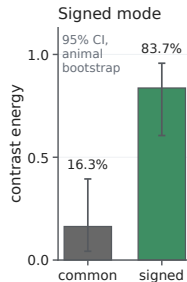
Animal teaching signals are signed and neuron-specific



opposite causal signs, fixed by the BCI mapping, predict opposite dendritic contrasts

83.7%

of contrast energy in the signed, neuron-specific mode — 6/6 animals



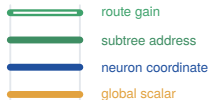
six-animal reanalysis of published source data (Francioni et al., Nature 2026); it does not resolve within-tree transport

The coordinate rung exists in vivo; transport is the open measurement.

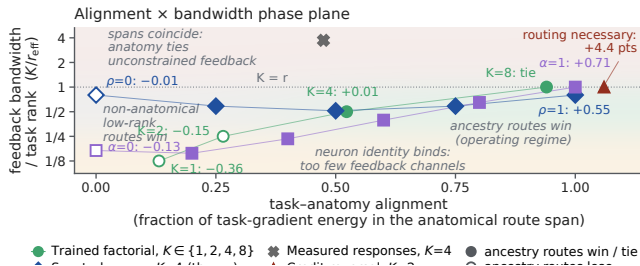
ACT V

What does this teach about learning in the brain?

One phase plane, four inequalities, wiring economics, testable predictions



Alignment $\times$ bandwidth decides when structure pays



- trained address factorial: interior win
- ◆ spectral sweep: alignment rotates the advantage in
- MICrONS controlled fields
- ✱ measured responses: spans coincide, predicted null
- ▲ credit reversal: routing becomes necessary
- open marker = anatomical routes lose

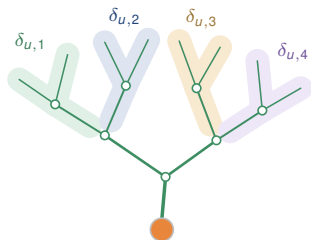
Alignment and bandwidth organize every result on one plane.

When a local rule wins is settled by four inequalities

- **CEILING** $M = I, \Sigma = 0$ nothing beats exact BP at matched step norm
- **NOISE** $\text{tr}[(I - P)\Sigma] > \|(I - P)\mathbf{g}\|_2^2$ beat stochastic BP by rejecting noise
- **SPAN** leading rank- K eigenspace $\subset$ ancestry span anatomy is Ky–Fan optimal only under alignment
- **ADDRESS** $\cos(\mathbf{g}^{\text{shared}}, \mathbf{g}^{\text{exact}}) = 1/\sqrt{1 + \text{CV}_w(\tilde{\alpha})^2} < 1$ addresses win once transport varies

Structure never beats the exact first-order ceiling; it pays only under alignment.

Why dendrites: one arbor multiplexes K channels



one neuron, K routed teaching channels



or: K point neurons, $K \times$ the feedback wiring

$$\text{rank}(\Phi_{T,K}) \leq K$$

A point system pays for the same addresses with routed coefficients, gated parameter groups, or extra units.

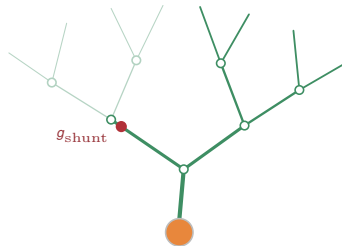
6.9%

of dense-oracle wiring buys 85% of its capture — $K=8$ ancestry channels, reconstructed cells

Dendrites relocate address resources; the ledger is wiring.

The theory is testable with focal dendritic inhibition

- PREDICTION 1
plasticity shifts below the branch
- PREDICTION 2
scales with dose and R_{in} ; saturates
- PREDICTION 3
needs high-conductance states
- PREDICTION 4
tracks capture, not morphology



focal dendritic inhibition

Branch-resolved plasticity separates credit routing from gain control.

Four questions, four answers

ACT II

One signed coordinate per neuron.

ACT III

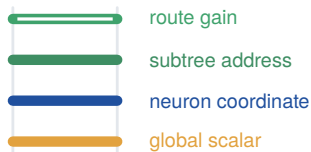
Eligibility $\times$ transported, routed error.

ACT IV

Yes — at the predicted budget, under alignment.

ACT V

One phase plane organizes every result.



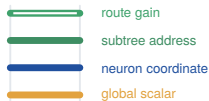
Thanks: Sabatini and Richards labs,
Kempner Institute compute and support.

Coordinates name neurons; trees address synapses; conductance tunes gain; alignment decides value.

BACKUP

Derivations, controls, audits

For questions



Appendix: the adjoint chain, from implicit function to path product

GENERAL ADJOINT

$$\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}, \quad J_V = \frac{\partial \mathbf{F}}{\partial \mathbf{V}}, \quad J_V^\top \mathbf{q} = \nabla_V \mathcal{L}$$

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^\top \frac{\partial \mathbf{F}}{\partial g_i} = x_i(E_i - V_n) q_n$$

synapse-local factor $\times$ compartment adjoint

In a reciprocal passive cable, $J_V = G$ is symmetric positive definite and $G^{-\top} = G^{-1}$: the adjoint is a Green's-function solve. For a general nonsymmetric or recurrent compartmental system the inverse transpose is essential.

This fixed-point adjoint is the steady-state member of the Almeida–Pineda recurrent-backpropagation lineage.

With the residual convention $\mathbf{F} = G\mathbf{V} - \mathbf{b}$. For the triangular tree system, $q_n = R_n^{\text{tot}} \partial \mathcal{L} / \partial V_n$, so the adjoint and compartment factorizations are the same identity with the input-resistance factor moved between terms.

Credit is always eligibility times an adjoint; only the adjoint changes.

Appendix: deriving $U(M)$ and its exactness caveat

DESCENT LEMMA

For L -Lipschitz $\nabla \mathcal{L}$ and $\mathbf{w}^+ = \mathbf{w} - \eta M \hat{\mathbf{g}}$, $\hat{\mathbf{g}} = \mathbf{g} + \xi$:

$$\begin{aligned} \mathbb{E} \mathcal{L}(\mathbf{w}^+) &\leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top M \mathbf{g} \\ &\quad + \frac{L\eta^2}{2} (\|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top)) \end{aligned}$$

Optimizing the right side over η when $\mathbf{g}^\top M \mathbf{g} > 0$:

$$U(M) = \frac{[\mathbf{g}^\top M \mathbf{g}]^2}{2L \{ \|M\mathbf{g}\|_2^2 + \text{tr}(M\Sigma M^\top) \}}$$

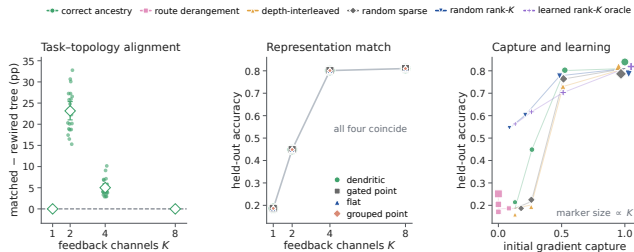
CAVEAT

The bound is an upper guarantee for a general L -smooth loss and is exact for the corresponding quadratic when curvature equals L on the update directions; comparisons of $U(M)$ otherwise order guarantees, not necessarily realized losses.

Diagnostics: $U(M)$ predicted norm-matched one-step progress with Spearman $\rho = 0.937$ and final held-out accuracy with $\rho = 0.916$ across 540 dendritic-route conditions; spectral capture alone predicted one-step progress at $\rho = 0.877$.

One-step geometry predicts within-family progress; trajectory-accrued advantages can escape it.

Appendix: factorial control battery and implementation equivalence



■ TOPOLOGY ALIGNMENT

matched tree minus rewired tree: $+0.231$ at $K=2$, $+0.050$ at $K=4$; 20/20 paired seeds each.

■ REPRESENTATION MATCH

dendritic, gated-point, flat and grouped-point implementations coincide exactly (maximum accuracy difference 0) when given identical routed fields.

■ CAPTURE IS NOT FATE

fields with similar initial capture but different route signs and task alignment learned differently.

An address resource any point system can buy with the same routes.

Appendix: input-validity audit and clean exact/BP agreement

OUTCOME-INDEPENDENT RULE

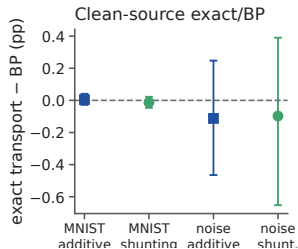
signed synthetic drive is invalid for a positive-conductance shunting denominator: 640 signed-shunting runs are invalid, and the 400-run inhibitory-dose family that depends on them is excluded with its 200 valid partners — 840 of 1,840 runs leave inference; all remain in the ledger.

CLEAN-SOURCE AUDIT

all 320 detached clean-source runs passed the configuration, log, metric and checkpoint audit; synthetic shunting uses a non-negative transfer.

AGREEMENT

all four depth-averaged exact-minus-BP intervals include zero; mean across 160 pairs -0.055 pp; largest single pair difference 5.90 pp.



10 seeds $\times$ 4 depths $\times$ 2 tasks $\times$ 2 cores $\times$ 2 methods — condition-average agreement, not pointwise identity

The clean audit supports agreement on average, never formal equivalence.

Appendix: reliability gain, and same-span conditioning

FINITE-STEP GAIN

For disjoint branches with signal S_b , noise N_b , step $\eta = c/L$:

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{c a_b^2}{2} (S_b + N_b) \right]$$

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}$$

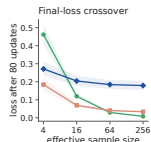
A point model supplied with identical branch gates must match — and did, to numerical precision.

trained branch model, 50 fresh seeds: aligned shunting beat the best global gain by 0.001817 in one-step loss reduction (49/50); final aligned-minus-no-shunt loss was null (0.000425, $P = 0.154$)

SAME-SPAN FIELD ITERATION

$$\mathbf{f}_{t+1} = \mathbf{f}_t - cA(\mathbf{f}_t - \mathbf{g} - \boldsymbol{\xi}_t)$$

$$\mathbb{E} \mathcal{L}_T = \frac{1}{2} \sum_j \left[B_{j,T} + \frac{\sigma^2}{n} V_{j,T} \right]$$



rank-eight same-span coordinates, condition numbers 1 / 15 / 388.5: slow modes filter noise at small n , orthogonal modes cut bias at large n ; predicted crossovers $n^* = 38.5, 8.25$

Conductance realizes the predicted ordering; conditioning trades bias against variance.

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All quantitative claims trace to the journal manuscript and its source-data package.